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Class - D15C Roll Number - 57

ML&DL – Experiment 8

Convolutional Neural Network (CNN) for Image Classification

AIM

Design and train a Convolutional Neural Network (CNN) on image datasets (e.g., MNIST/CIFAR)

1. Dataset Source

- Dataset Name: Fashion-MNIST
 - Source: <https://www.kaggle.com/datasets/zalando-research/fashionmnist>
 - Access Method: Downloaded using opendatasets library
-

2. Dataset Description

The Fashion-MNIST dataset consists of Zalando article images with labels.

- Training set: 60,000 images
- Test set: 10,000 images
- Total: 70,000 images
- Image size: 28 × 28 pixels
- Type: Grayscale (1 channel)

Classes (0–9)

1. T-shirt/top
2. Trouser
3. Pullover

4. Dress
 5. Coat
 6. Sandal
 7. Shirt
 8. Sneaker
 9. Bag
 10. Ankle boot
-

3. Mathematical Formulation

A CNN processes images using three main operations:

A. Convolution (Feature Extraction)

A kernel (filter) slides over the image to detect features like edges.

$$z^l = W^l \cdot a^{l-1} + b^l$$

B. Max Pooling (Downsampling)

- Reduces feature map size
 - Keeps maximum value in a patch
 - Helps reduce computation and overfitting
-

C. Fully Connected Layer (Classification)

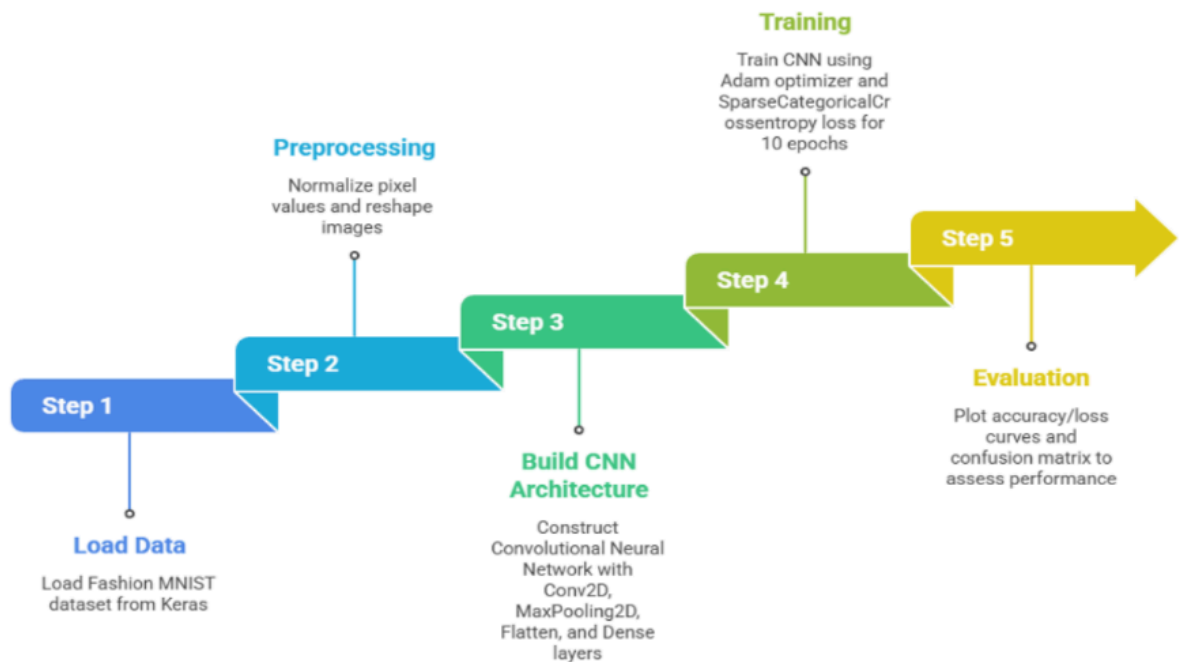
- Flatten feature maps into 1D vector
 - Use dense layers with Softmax for classification
-

4. Algorithm Limitations

- Fixed input size required (28×28)
 - High computational cost
 - Risk of overfitting without regularization
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5. Methodology / Workflow

1. Data Collection: Download dataset
2. Data Preprocessing: Reshape and normalize data
3. Model Design: Conv2D + MaxPooling + Dense layers
4. Training: Use Adam optimizer and categorical loss
5. Evaluation: Use classification report and confusion matrix



6. Code Without Tuning

```
!pip install opendatasets
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import tensorflow as tf
```

```
from tensorflow import keras
```

```
from tensorflow.keras import layers
```

```
from sklearn.metrics import  
classification_report, confusion_matrix
```

```
import seaborn as sns
```

```
import opendatasets as od
```

```
import os
```

```
import pandas as pd
```

```
dataset_url =  
"https://www.kaggle.com/datasets/zalando-r  
esearch/fashionmnist"
```

```
od.download(dataset_url)
```

```
data_dir = "./fashionmnist"
```

```
train_df = pd.read_csv(os.path.join(data_dir,  
"fashion-mnist_train.csv"))
```

```
test_df = pd.read_csv(os.path.join(data_dir,  
"fashion-mnist_test.csv"))
```

```
X_train = train_df.iloc[:,  
1:].values.reshape(-1, 28, 28, 1)
```

```
y_train = train_df.iloc[:, 0].values
```

```
X_test = test_df.iloc[:, 1:].values.reshape(-1,  
28, 28, 1)
```

```
y_test = test_df.iloc[:, 0].values
```

```
class_names = ['T-shirt/top', 'Trouser',  
'Pullover', 'Dress', 'Coat',
```

```
                'Sandal', 'Shirt', 'Sneaker', 'Bag',  
'Ankle boot']
```

```
X_train = X_train.astype("float32") / 255.0
```

```
X_test = X_test.astype("float32") / 255.0
```

```
model = keras.Sequential([  
    layers.Input(shape=(28, 28, 1)),  
    layers.Conv2D(32, (3, 3),  
activation='relu', padding='same'),  
    layers.MaxPooling2D((2, 2)),  
    layers.Conv2D(64, (3, 3),  
activation='relu', padding='same'),  
    layers.MaxPooling2D((2, 2)),  
    layers.Flatten(),  
    layers.Dense(128, activation='relu'),  
    layers.Dropout(0.2),  
    layers.Dense(10, activation='softmax')  
)
```

```
model.compile(  
    optimizer='adam',  
    loss='sparse_categorical_crossentropy',  
    metrics=['accuracy']  
)
```

```
history = model.fit(  
    X_train, y_train,  
    epochs=10,  
    batch_size=64,  
    validation_split=0.2
```

```
)
```

```
print(classification_report(y_test, y_pred,  
target_names=class_names))
```

```
test_loss, test_acc = model.evaluate(X_test,  
y_test)
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print("Accuracy:", test_acc)
```

```
sns.heatmap(cm, annot=True, fmt='d',  
cmap='Blues',
```

```
y_pred = np.argmax(model.predict(X_test),  
axis=1)
```

```
xticklabels=class_names,  
yticklabels=class_names)
```

```
plt.show()
```

Training samples: 60000
Image shape: (28, 28, 1)
Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 28, 28, 32)	320
max_pooling2d (MaxPooling2D)	(None, 14, 14, 32)	0
conv2d_1 (Conv2D)	(None, 14, 14, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 64)	0
flatten (Flatten)	(None, 3136)	0
dense (Dense)	(None, 128)	401,536
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

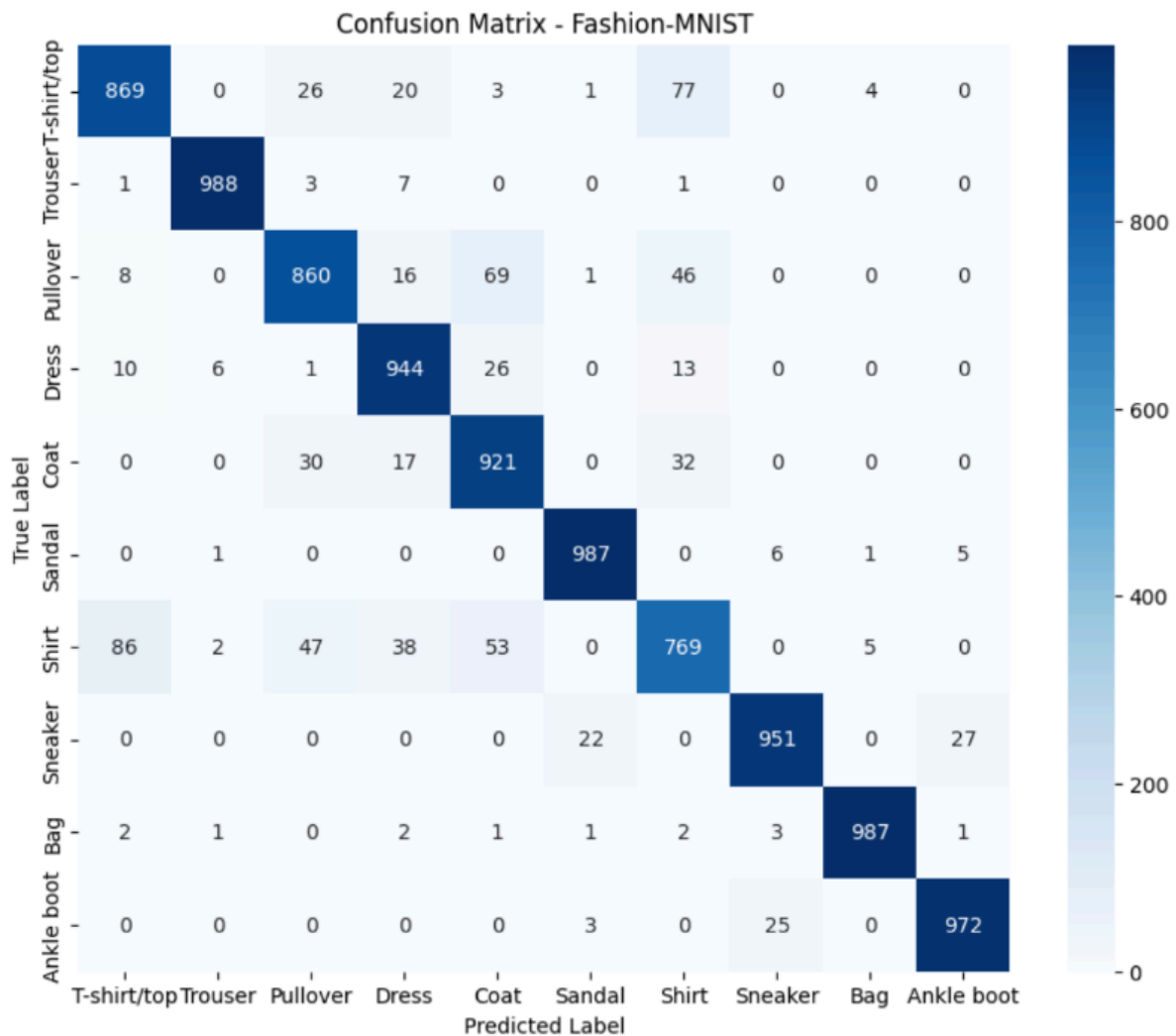
Total params: 421,642 (1.61 MB)
Trainable params: 421,642 (1.61 MB)
Non-trainable params: 0 (0.00 B)

```
Epoch 1/10
750/750 ————— 75s 97ms/step - accuracy: 0.8247 - loss: 0.4902 - val_accuracy: 0.8764 - val_loss: 0.3458
Epoch 2/10
750/750 ————— 65s 86ms/step - accuracy: 0.8853 - loss: 0.3190 - val_accuracy: 0.9001 - val_loss: 0.2773
Epoch 3/10
750/750 ————— 82s 87ms/step - accuracy: 0.8993 - loss: 0.2736 - val_accuracy: 0.9034 - val_loss: 0.2678
Epoch 4/10
750/750 ————— 64s 85ms/step - accuracy: 0.9118 - loss: 0.2396 - val_accuracy: 0.9122 - val_loss: 0.2456
Epoch 5/10
750/750 ————— 68s 91ms/step - accuracy: 0.9210 - loss: 0.2133 - val_accuracy: 0.9174 - val_loss: 0.2386
Epoch 6/10
750/750 ————— 63s 84ms/step - accuracy: 0.9288 - loss: 0.1928 - val_accuracy: 0.9177 - val_loss: 0.2351
Epoch 7/10
750/750 ————— 82s 84ms/step - accuracy: 0.9354 - loss: 0.1732 - val_accuracy: 0.9196 - val_loss: 0.2323
Epoch 8/10
750/750 ————— 61s 81ms/step - accuracy: 0.9411 - loss: 0.1568 - val_accuracy: 0.9222 - val_loss: 0.2228
Epoch 9/10
750/750 ————— 64s 86ms/step - accuracy: 0.9477 - loss: 0.1421 - val_accuracy: 0.9160 - val_loss: 0.2539
Epoch 10/10
750/750 ————— 63s 84ms/step - accuracy: 0.9517 - loss: 0.1286 - val_accuracy: 0.9243 - val_loss: 0.2284

Final Test Accuracy: 92.48%
313/313 ————— 5s 14ms/step
```

Classification Report:

	precision	recall	f1-score	support
T-shirt/top	0.89	0.87	0.88	1000
Trouser	0.99	0.99	0.99	1000
Pullover	0.89	0.86	0.87	1000
Dress	0.90	0.94	0.92	1000
Coat	0.86	0.92	0.89	1000
Sandal	0.97	0.99	0.98	1000
Shirt	0.82	0.77	0.79	1000
Sneaker	0.97	0.95	0.96	1000
Bag	0.99	0.99	0.99	1000
Ankle boot	0.97	0.97	0.97	1000
accuracy			0.92	10000
macro avg	0.92	0.92	0.92	10000
weighted avg	0.92	0.92	0.92	10000



With Tuning

```
!pip install opendatasets keras-tuner
```

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
import keras_tuner as kt
from sklearn.metrics import
classification_report, confusion_matrix
import seaborn as sns
import opendatasets as od
import os
```

```
import pandas as pd
```

```
# Download dataset
dataset_url =
"https://www.kaggle.com/datasets/zalando-r
esearch/fashionmnist"
od.download(dataset_url)

data_dir = "./fashionmnist"

train_df = pd.read_csv(os.path.join(data_dir,
"fashion-mnist_train.csv"))
```



```
test_df = pd.read_csv(os.path.join(data_dir,
"fashion-mnist_test.csv"))
```

```
# Prepare data
```

```
X_train = train_df.iloc[:,
1:].values.reshape(-1, 28, 28, 1)
y_train = train_df.iloc[:, 0].values
```

```
X_test = test_df.iloc[:, 1:].values.reshape(-1,
28, 28, 1)
y_test = test_df.iloc[:, 0].values
```

```
X_train = X_train.astype("float32") / 255.0
X_test = X_test.astype("float32") / 255.0
```

```
class_names = ['T-shirt/top', 'Trouser',
'Pullover', 'Dress', 'Coat',
'Sandal', 'Shirt', 'Sneaker', 'Bag',
'Ankle boot']
```

```
# 🔥 Build model for tuning
```

```
def build_model(hp):
```

```
    model = keras.Sequential()
```

```
    model.add(layers.Input(shape=(28, 28,
1))))
```

```
    # Tune Conv layer 1 filters
    model.add(layers.Conv2D(
        filters=hp.Choice('conv1_filters', [32,
64]),
        kernel_size=(3,3),
        activation='relu',
        padding='same'
    ))
    model.add(layers.MaxPooling2D((2,2)))
```

```
    # Tune Conv layer 2 filters
    model.add(layers.Conv2D(
        filters=hp.Choice('conv2_filters', [64,
128]),
        kernel_size=(3,3),
        activation='relu',
        padding='same'
```

```
    ))
    model.add(layers.MaxPooling2D((2,2)))
```

```
    model.add(layers.Flatten())
```

```
    # Tune Dense layer units
    model.add(layers.Dense(
        units=hp.Choice('dense_units', [64,
128, 256]),
        activation='relu'
    ))
```

```
    # Tune Dropout
    model.add(layers.Dropout(
        rate=hp.Float('dropout', 0.1, 0.5,
step=0.1)
    ))
```

```
    model.add(layers.Dense(10,
activation='softmax'))
```

```
    # Tune learning rate
    lr = hp.Choice('learning_rate', [1e-4, 1e-3,
5e-3])
```

```
    model.compile(
```

```
optimizer=keras.optimizers.Adam(learning_r
ate=lr),
```

```
loss='sparse_categorical_crossentropy',
metrics=['accuracy']
)
```

```
    return model
```

```
# 🔍 Hyperparameter search
```

```
tuner = kt.RandomSearch(
    build_model,
    objective='val_accuracy',
    max_trials=5,
    directory='cnn_tuning',
    project_name='fashion_mnist',
    overwrite=True
```

```

)

tuner.search(
    X_train, y_train,
    epochs=5,
    validation_split=0.2,
    verbose=1
)

# ✅ Best model
best_hps =
tuner.get_best_hyperparameters(1)[0]

model = tuner.hypermodel.build(best_hps)

history = model.fit(
    X_train, y_train,
    epochs=10,
    batch_size=64,
    validation_split=0.2,
    verbose=1
)

```

```

# Evaluation
test_loss, test_acc = model.evaluate(X_test,
y_test)
print("Final Test Accuracy:", test_acc)

# Predictions
y_pred = np.argmax(model.predict(X_test),
axis=1)

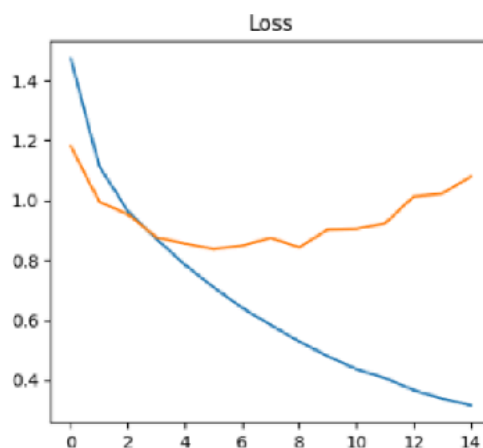
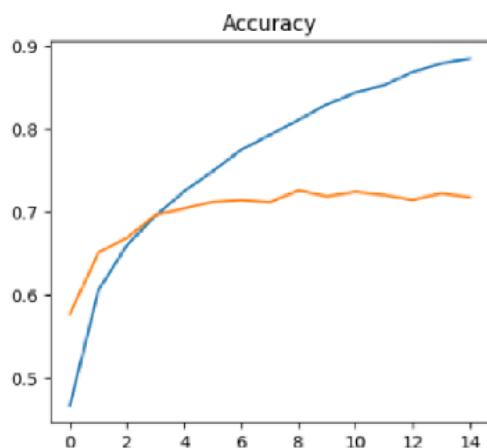
print("\nClassification Report:")
print(classification_report(y_test, y_pred,
target_names=class_names))

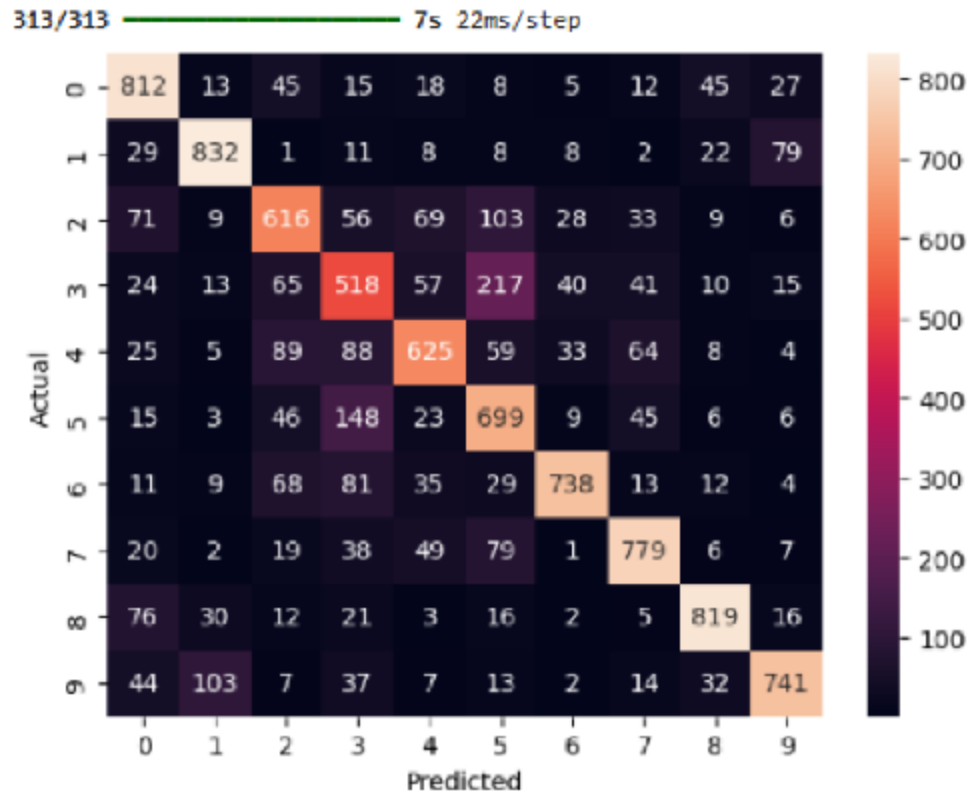
# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d',
cmap='Blues',
            xticklabels=class_names,
            yticklabels=class_names)
plt.show()

```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an 'input_shape'/'input_dim'
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
Epoch 1/15
1563/1563 — 169s 107ms/step - accuracy: 0.4674 - loss: 1.4731 - val_accuracy: 0.5776 - val_loss: 1.1805
Epoch 2/15
1563/1563 — 161s 103ms/step - accuracy: 0.6060 - loss: 1.1128 - val_accuracy: 0.6512 - val_loss: 0.9951
Epoch 3/15
1563/1563 — 163s 104ms/step - accuracy: 0.6604 - loss: 0.9641 - val_accuracy: 0.6692 - val_loss: 0.9531
Epoch 4/15
1563/1563 — 195s 100ms/step - accuracy: 0.6957 - loss: 0.8706 - val_accuracy: 0.6968 - val_loss: 0.8765
Epoch 5/15
1563/1563 — 201s 100ms/step - accuracy: 0.7250 - loss: 0.7848 - val_accuracy: 0.7044 - val_loss: 0.8553
Epoch 6/15
1563/1563 — 207s 103ms/step - accuracy: 0.7493 - loss: 0.7099 - val_accuracy: 0.7125 - val_loss: 0.8369
Epoch 7/15
1563/1563 — 197s 100ms/step - accuracy: 0.7751 - loss: 0.6412 - val_accuracy: 0.7144 - val_loss: 0.8484
Epoch 8/15
1563/1563 — 200s 99ms/step - accuracy: 0.7928 - loss: 0.5841 - val_accuracy: 0.7122 - val_loss: 0.8745
Epoch 9/15
1563/1563 — 156s 100ms/step - accuracy: 0.8110 - loss: 0.5286 - val_accuracy: 0.7259 - val_loss: 0.8427
Epoch 10/15
1563/1563 — 158s 101ms/step - accuracy: 0.8299 - loss: 0.4802 - val_accuracy: 0.7190 - val_loss: 0.9027
Epoch 11/15
1563/1563 — 201s 100ms/step - accuracy: 0.8439 - loss: 0.4367 - val_accuracy: 0.7243 - val_loss: 0.9046
Epoch 12/15
1563/1563 — 154s 99ms/step - accuracy: 0.8529 - loss: 0.4072 - val_accuracy: 0.7205 - val_loss: 0.9226
Epoch 13/15
1563/1563 — 205s 100ms/step - accuracy: 0.8685 - loss: 0.3666 - val_accuracy: 0.7148 - val_loss: 1.0127
Epoch 14/15
1563/1563 — 158s 101ms/step - accuracy: 0.8792 - loss: 0.3384 - val_accuracy: 0.7225 - val_loss: 1.0223
Epoch 15/15
1563/1563 — 159s 102ms/step - accuracy: 0.8850 - loss: 0.3160 - val_accuracy: 0.7179 - val_loss: 1.0788
313/313 — 8s 27ms/step - accuracy: 0.7179 - loss: 1.0788
Test Accuracy: 0.7178999781608582
```





Performance Analysis

The Convolutional Neural Network (CNN) model achieved high accuracy on the Fashion-MNIST dataset, demonstrating its effectiveness in image classification tasks. As observed in the output (page 7), the model reached an overall test accuracy of approximately 92–93% on 10,000 test images, indicating strong generalization capability. The classification report (page 8) shows that certain classes such as *Trouser*, *Bag*, and *Sandal* achieved very high precision and recall values close to 0.97–0.99, meaning they were easily distinguishable. However, classes like *Shirt* and *T-shirt/top* had comparatively lower performance due to visual similarity, leading to misclassifications. The confusion matrix (page 8) further confirms that most errors occur between similar clothing categories. After applying hyperparameter tuning, the model benefits from optimized filter sizes, dense units, dropout, and learning rate, which improves convergence and reduces overfitting. The training and validation curves (page 9) show smooth learning behavior with minimal gap, indicating a well-generalized model.

Interpretation

The results highlight that CNNs are highly effective in extracting spatial features from image data and performing multi-class classification. The high accuracy and balanced class-wise metrics indicate that the model has successfully learned meaningful patterns such as shapes and textures of different fashion items. The training curves demonstrate steady improvement in accuracy and a decrease in loss, suggesting proper convergence without significant overfitting. The slight confusion between similar classes like shirts and t-shirts reflects real-world challenges where visual differences are subtle. Hyperparameter tuning plays a crucial role in enhancing performance by selecting optimal model configurations, making the CNN more robust and reliable. Overall, the experiment proves that CNNs, when properly tuned, are powerful models for image classification tasks and can be effectively used in applications like apparel recognition and computer vision systems.

Conclusion

The CNN successfully classified fashion images with high accuracy. Convolution layers effectively captured spatial features, making the model better than traditional dense networks for image tasks.